AeroTwin

Predictive clinical intelligence and cardiorespiratory risk stratification powered by environmental data analytics.

TEAM NAME

Binary-Minds

TEAM MEMBERS

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Problem Statement & Proposed Solution

The Problem

Reactive Smog Treatment

Hospitals treat cardiorespiratory distress only after irreversible exposure to severe winter PM2.5 occurs.

Opaque Risk Factors

Patients lack an accessible way to understand how specific pollution levels impact their Heart Rate and SpO2.

Lack of Actionable Guidelines

Clinical staff cannot easily quantify the exact physiological benefit of an N95 mask or staying indoors.

AeroTwin Solution

Automated 24h Risk Forecaster

An XGBoost engine scoring 24-hour flare-up risk directly from CPCB PM2.5 data and wearable telemetry.

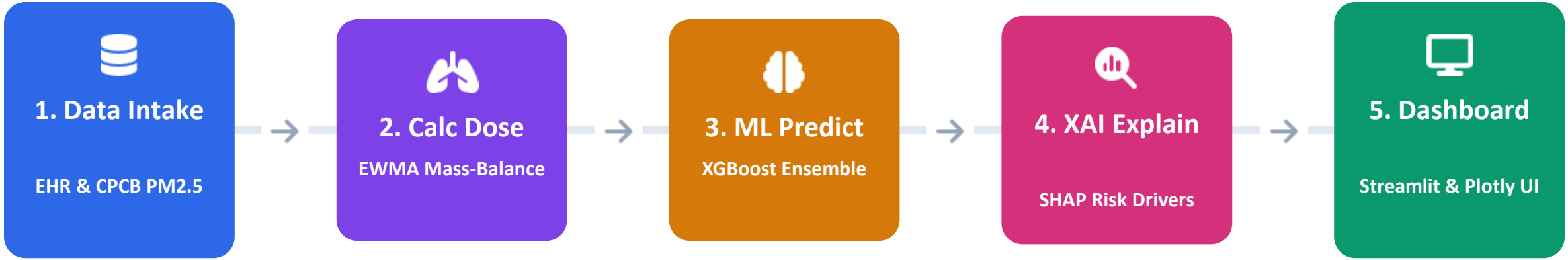
SHAP-Powered Explainability

Breaks down primary contributing factors (e.g., Asthma history, PM2.5 surge) in clear clinical terms.

Interactive Simulator

Dashboards proving how behavioral interventions mathematically lower internal particle dose.

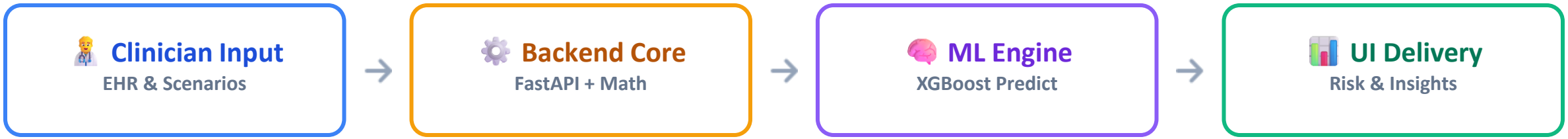
Technical Flow & Infrastructure



Backend & Model	Python 3.12, Pandas, XGBoost Ensemble
Intelligence (XAI)	SHAP Values (Feature contributions & impact)
Frontend UI	Streamlit with Plotly Graphing Engine & FastAPI

System Architecture: High vs Low Level

HIGH LEVEL DESIGN (HLD)



LOW LEVEL DESIGN (LLD) PIPELINE

01 **Intake Module**
Age, Asthma, HTN, Current SpO2, Hourly PM2.5 Series

02 **Validation & Transform**
Null checks, imputation, and Personal Exposure Math

03 **Inference Module**
XGBoost probability scoring & SHAP feature contributions

Data Flow Diagram (DFD): Level 0 & Level 1

DFD Level 0

CONTEXT

Hospital Staff / Clinical Analyst

Primary System Operator

↓ Input: Patient Data & Vitals Output: Risk Results ↑

0

Healthcare Prediction System

Central Processing Unit & ML Inference Engine

↓ Analytics Data & Insights Report Requests ↑

Clinical Dashboard & Reports

Visualization & Export Interface

DFD Level 1: Modular Decomposition

DECOMPOSED

Staff

Sends: Patient Details

PROCESS 1.0

Manage Patient Data

Intake, demographics & validation

Patient

Provides: Medical History

↓ Verified Patient Pipeline ↓

Staff Query

Gets: Charts & Insights

PROCESS 2.0

Generate Analytics & Reports

ML risk scoring & factor explanations

Dashboard

Receives: Visual Metrics

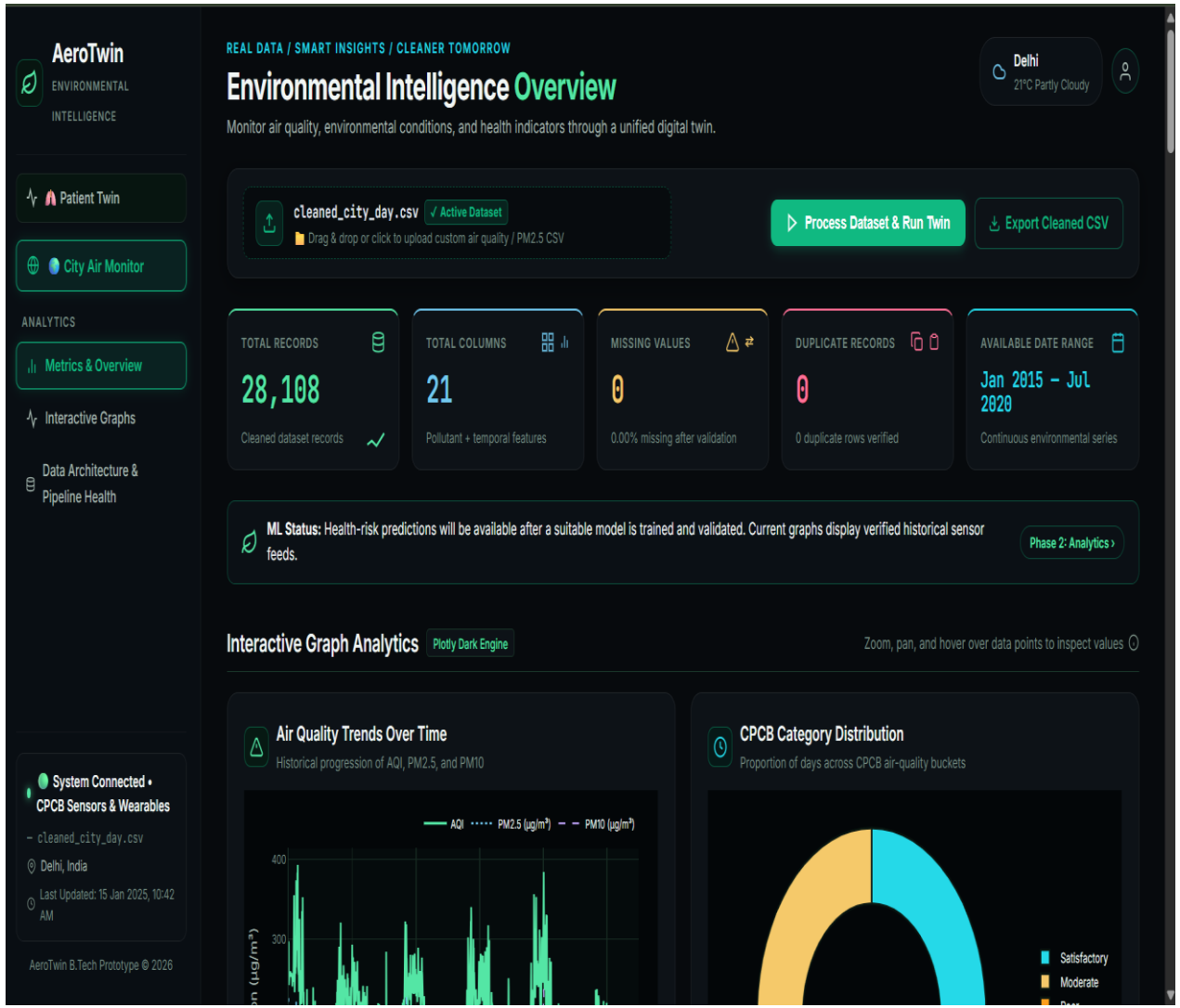
STORE D1 **Healthcare Database**

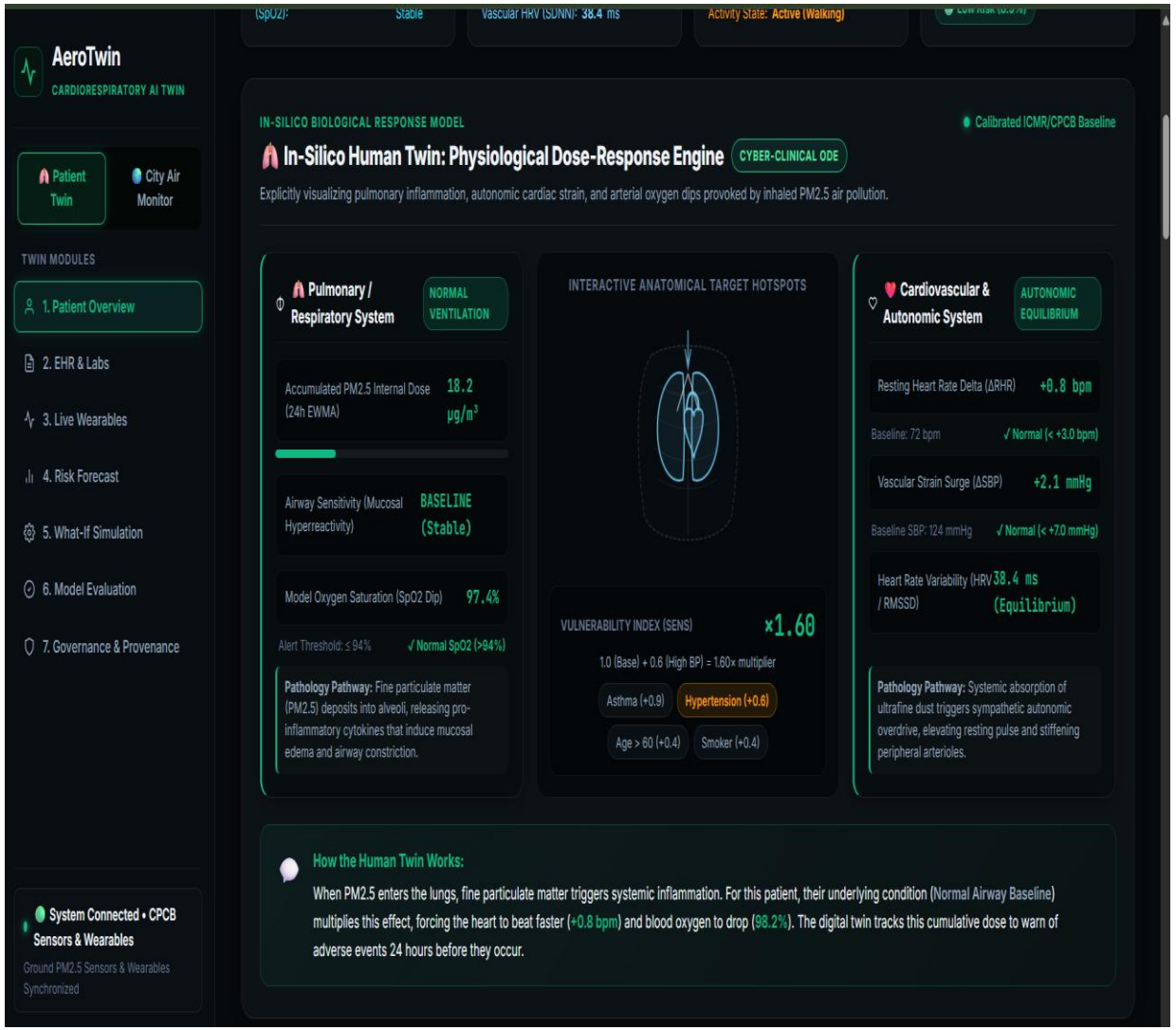
Stores patient demographics, medical episodes, and inference logs.

PERSISTENT

Twin Intelligence Dashboards

Interactive Analytics Interface





Impacts & Benefits

Early Risk Identification

Flags high-risk cardiorespiratory events 24 hours in advance for preventative care.

Explainable AI

SHAP explanations show exactly which physical factors drive each score.

Population Insight

Dashboards reveal respiratory strain patterns by age, diagnosis, and AQI.

Treatment Simulator

What-if scenario testing lets users compare how N95 masks alter predicted risk.

Privacy by Design

Built strictly on public CPCB datasets and synthetic patient EHR (HIPAA/GDPR).

Reproducible Stack

Open-source Python stack and modular codebase makes the system easy to retrain.

References & Research

PRIMARY DATASETS

CPCB Air Quality Portal

Central Control Room portal ground-station PM2.5 readings.

Synthea Generator

Synthetic health records used for cohort modeling.

LIBRARIES & DOCS

Core ML Ecosystem

Official documentation for scikit-learn, XGBoost, and Pandas.

SHAP Architecture

Game-theoretic approaches to explain machine learning outputs.

GOVERNANCE & SPECS

Project Design Guide

AeroTwin Digital Twin architectural specifications.

Evaluation Matrix

Group-Shuffle Split methodology ensuring zero data leakage.